

CHIRAG GUPTA

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[LeetCode](#) | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

EDUCATION

MCA (AI & ML)

KR Mangalam University, India

2025 – 2027

CGPA: 9.08

OPEN SOURCE & ACHIEVEMENTS

- **hypersistence-utils** – Contributed a **merged pull request (PR #859)** to an open-source **Hibernate / JPA** utility library used across production Java projects.
- Solved **400+ Data Structures & Algorithms problems** on [Leetcode](#) and [GFG](#)

PROJECTS

SIVP - Multi-Agent AI Startup Validation Platform

July 2026

Tools: Java, Spring Boot, React.js, Supabase (PostgreSQL), n8n, Google Gemini, REST APIs, OAuth 2.0, Row-Level Security, Docker, AWS(Elastic IP), GitHub Actions, Maven [\[GitHub\]](#) [\[Deployed Link\]](#)

- Parallelized the 7-agent analysis stage, reducing pipeline runtime by **75% (4min → 1min)** via concurrent agent execution.
- Implemented **request-level caching** with content-hash keys, serving duplicate idea submissions in **near-instant (<1s)** response time instead of re-running the pipeline.
- Added a **pre-flight validation gate**, rejecting invalid ideas in **under 5 seconds** and cutting wasted compute to **zero** before they reach the pipeline.
- Achieved **100% pipeline uptime** under partial failures via circuit-breaker-style fault isolation, replacing a single-point-of-failure response handler.
- Deployed internally for the college's incubation center(KEIC), enabling mentor-review workflows where submitted startup ideas are tracked, scored, and reviewed by mentors across cohorts in one system.

Home4Paws - Dog Adoption Platform

June 2026

Tools: Spring Boot, Java, React.js, PostgreSQL, Redis, REST APIs, JWT, OAuth 2.0, Spring Security, JPA/Hibernate, Server-Sent Events, Razorpay, Docker, Nginx, AWS EC2, Maven [\[GitHub\]](#) [\[Deployed Link\]](#)

- Designed a **Spring Boot backend** with 12 entities across 8 REST controllers (40 endpoints), covering pet adoption, e-commerce checkout, subscriptions, and real-time chat in one unified API layer.
- Built a **custom JWT validation filter** that verifies Supabase-issued tokens at the API layer, extracting user identity/role and syncing it into a separate Postgres user store — decoupling authentication (Supabase) from authorization (own role-based DB).
- Built **real-time chat using SSE + Redis Pub/Sub for cross-instance message fan-out**, solving the classic distributed-systems problem where SSE breaks when sender and receiver land on different backend instances behind a load balancer.
- Integrated **Razorpay payments** across one-time orders and recurring subscriptions, unifying two billing flows through a shared, idempotent payment service layer.
- Added **middleware-based request observability** via a custom logging filter, capturing full API traffic without touching individual controllers.

NanoBase – Mini Database Engine

February 2026 – April 2026

Tools: Core Java, B-Tree, File I/O, Java Serialization, SQL Parser, Lexer, Recursive Descent Parsing, Generics, REPL [\[GitHub\]](#) [\[Demo\]](#)

- Architected the write path to eliminate $O(n^2)$ I/O amplification, cutting bulk-write time from **210s to 13s (16x faster)** on 12,000-record workloads.
- Built a query-planning fix that routes indexed database lookups through the index instead of falling back to full table scans, reducing lookup latency from **2.4ms to 0.01ms (220x) at 20K+ records**
- Built crash-safe persistence using atomic file commits and replayable write-ahead logs, verified via live process-kill fault injection with **zero data corruption** across repeated trials.
- Enabled safe concurrent access across multiple simultaneous threads using fine-grained locking, validated under load with **16,000+ concurrent writes and zero race conditions**.

INTERNSHIP EXPERIENCE

Data Analyst Intern – Edulyt India - [\[Validate\]](#)

June 2024 – August 2024

- Analyzed **100,000+ financial records** for credit risk assessment, performing **end-to-end data mapping** across 3+ source systems.
- Validated and cleaned datasets using **SQL queries**, resolving **20%+ data quality issues** including nulls, duplicates, and format inconsistencies across **6+ Agile sprint cycles**.

Data Science Intern – Coratia Technologies

October 2023 – December 2023

- Prepared, cleaned, and validated **10,000+ records** across **4+ data sources**.
- Performed **Exploratory Data Analysis (EDA)** on large datasets to identify trends, anomalies, missing values, and inconsistencies, uncovering and resolving **500+ data quality issues** and reducing downstream errors by **40%**.

SKILLS

- Languages: Java, SQL.
- Backend: Spring Boot, Spring Data JPA, Hibernate, REST APIs, JWT, OAuth 2.0, Microservices.
- Data Stores: PostgreSQL, Supabase, DBEaver, Redis (caching, Pub/Sub).
- Infrastructure: Docker, AWS (EC2, Elastic IP), Multi-Agent Orchestration (n8n), GitHub Actions (CI/CD), Git, Maven, Postman.
- Systems: DBMS, Data Structures & Algorithms, OOP, Operating Systems, Agile.